HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and executed test plans for prototype medical device performance evaluations, analyzing results to identify optimal solutions to technical challenges and thoroughly documenting findings.
- Translated design inputs into detailed specifications and test plans for novel medical devices, ensuring alignment with user needs and project objectives.
- Identified and selected appropriate product materials and assembly methods, defining process settings to meet performance and cost targets for new device development.
- Contributed to a project team focused on the early phase of product development and testing for a medical device start-up, meeting key performance and schedule objectives.
- Assisted in developing product training materials by understanding physician techniques and clinical needs for a new treatment area.
- Liaised with external vendors and resources to achieve project goals, ensuring timely delivery of critical components and services for device development.
- Containerized core application services using **Docker** and deployed them to an **AWS EKS** cluster, configuring basic resource limits for predictable rollout and performance.

SKILLS

Engineering – R&D; Engineering, Medical Device Development, Neuro-Interventional Devices, Design Inputs, Test Plans, Prototype Evaluation, Material Selection, Assembly Methods, Process Settings, Clinical Needs

Software Development – Java, Spring Boot, REST APIs, Python, SQL, PostgreSQL, MySQL, Oracle, Docker, Git, Jenkins

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Linux Shell Scripting

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, R, pandas, NumPy, scikit-learn, Statistical Analysis

Project Management & Tools – Jira, Confluence, Agile, Scrum, Change Management

PROJECTS

Medical Device Prototype Testing Framework

- Designed and implemented a Python-based framework for automating performance testing of early-stage medical device prototypes, integrating with sensor data acquisition systems.
- Developed detailed test protocols and analysis scripts to evaluate prototype performance against design specifications, identifying failure modes and areas for iterative improvement.
- Documented test results and provided actionable insights to the R&D; team, contributing to the refinement of device design and material selection processes.

Component Integration for Enterprise Systems

- Engineered a **Java Spring Boot** microservice to integrate with an existing ERP system, facilitating data exchange for operational reporting.
- Developed **REST APIs** for seamless communication between the new service and downstream applications, ensuring data consistency across enterprise platforms.
- Wrote comprehensive unit tests using **JUnit**, achieving over 85% code coverage for the critical integration components.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)